Centroid Geometry and Exact Formulations in `c2blue55` (`v8.2`)

512D Prior Centering, Orthonormal Rademacher-Hadamard 1-Bit RaBitQ, Conformal Radius, INT8 Heads, and Offline CUSUM Guards

Engineering Principle (Overview)

On high-throughput DNS sockets or host telemetry rings, running a deep neural network per packet exceeds microsecond latency budgets and triggers unpredictable GC pauses through heap allocations. `c2blue55` replaces heavy runtime inference with a **zero-allocation deterministic cascade** (`0 B/op`, `6.68–11.94 µs/op` with `GOEXPERIMENT=simd` / `8.63–16.32 µs/op` scalar ceiling, standalone stripped `c2agent` binary of `7.58 MiB`) backed by read-only `mmap`-projected vector floppies (`.c2book`, format `C2FLOP1`).

Every observation (up to `4,096` bytes) is mapped into `512 float32` coordinates, centered against an analytical neutral point, rotated isometrically by a Fast Walsh-Hadamard Transform (`FHT512`), and compressed into `512` sign bits (`64` bytes). Comparing a live event against a threat catalog reduces to `8` hardware `XOR` + `POPCNT` instructions on `64`-bit words.

Across the `c2blue55` codebase, the term **centroid** refers to four distinct, chained mathematical objects:

Mathematical Object	Representation	Functional Role	Source Implementation
1. Prior Reference μ_{ref}	Fixed `[512]float32` array	Subtracts uniform-byte expectation ($x = x_{\text{raw}} - \mu_{\text{ref}}$)	`internal/engine/feature_extractor.go`
2. RaBitQ Threat Centroids	`[8]uint64` (512 bits) in `CodebookEntry` (72 B)	`L1b` threat veto via $d_H(q,c) \le R_{\text{block}}$ (+ `L1a` spread centroid)	`rabitq512.go`, `floppy_engine.go`, `rabitq_probe.go`
3. Empirical Centroids μ_B, μ_H	`[512]float32` means quantized to `[512]int8`	Linear separating hyperplane ($W \propto \mu_H - \mu_B$) in `L1b`	`cmd/c2forge/floppy_builder.go`
4. Tracked Baseline μ_t	Offline `[512]float32` sequence	Anti-poisoning Euclidean + Page's `CUSUM` recertification gate	`internal/engine/drift_guard.go`

1. 512D Feature Projection and Prior Reference Centroid

When evaluating an event `Probe_event_t` on the three general tournament floppies (`floppy_lolbas.c2book`, `floppy_dns_c2.c2book`, `floppy_agent_mcp.c2book`), `FeatureExtractor` resolves the full payload up to $\text{PageSize} = 4\,096$ bytes without stopping on interior `NUL` bytes, populating a raw vector $x_{\text{raw}} \in \mathbb{R}^{512}$ partitioned into seven contiguous regions:

$$\dim(x_{\text{raw}}) = \underbrace{256}_{\text{unigrams}} + \underbrace{128}_{\text{bigrams}} + \underbrace{64}_{\text{trigrams}} + \underbrace{16}_{\text{local entropies}} + \underbrace{12}_{\text{ASCII classes}} + \underbrace{8}_{\text{header}} + \underbrace{28}_{\text{aggregates}} = 512$$

To prevent static baseline offsets from dominating sign quantization, `FeatureExtractor.center` subtracts the compile-time **prior reference centroid** $\mu_{\text{ref}} \in \mathbb{R}^{512}$ (`buildReferenceCentroid`), defined analytically as the expected value of a uniform byte distribution over $\{0, \ldots, 255\}$:

$$x = x_{\text{raw}} - \mu_{\text{ref}}, \qquad \mu_{\text{ref}}[i] = \begin{cases} \frac{1}{256} & \text{if } i \in [0, 255] \quad \text{(uniform byte unigram frequencies)} \\ \frac{1}{128} & \text{if } i \in [256, 383] \quad \text{(2-byte shift-XOR bigram buckets)} \\ \frac{1}{64} & \text{if } i \in [384, 447] \quad \text{(3-byte multiplicative hash buckets)} \\ 1.0 & \text{if } i \in [448, 463] \quad \text{(16 sliding entropy windows normalized by 8 bits)} \\ \frac{\lvert C_{i-464} \rvert}{256} & \text{if } i \in [464, 475] \quad \text{(cardinalities of the 12 character classes)} \\ 0 & \text{if } i \in [476, 511] \quad \text{(8 header fields and 28 statistical aggregates)} \end{cases}$$

where the 12 class cardinalities $|C_k|$ are `10` (vowels), `42` (consonants), `26` (uppercase), `26` (lowercase), `10` (digits), `12` (hex letters), `32` (punctuation), `2` (space/tab), `33` (control), `128` (bytes $\ge$ 0x80), `95` (printable ASCII), and `62` (alphanumeric). *(Note: Cold-Family Split DNS floppies `floppy_dns_tunnel.c2book` and `floppy_dns_dga.c2book` use parent-blind domain extractors `ExtractDNSTunnel512` and `ExtractDNSDGA512` that center structural DNS features directly into $[-1, 1]^{512}$ before feeding `QuantizeFHT512`.)*

2. 1-Bit RaBitQ FHT512 Threat Centroids and Veto Radius

Inside a `.c2book` floppy, every real attack vector from the training partition (plus any uncovered out-of-distribution motif isolated by `DiscoverRedhostPatterns`) is compiled into a 72-byte `CodebookEntry`: 64-byte `[8]uint64` bitcode, `ThreatID` (`uint32`), `Subsystem` (`uint16`), and `Severity` (`uint16`).

2.1 Orthonormal Rademacher–Hadamard Rotation (`ProjectRabitq`)

Direct 1-bit quantization along canonical axes collapses sparse spikes concentrated on a handful of coordinates. `QuantizeFHT512` (`rabitq512.go`) applies a structured orthonormal rotation without allocating a dense 512×512 matrix:

$$p = \frac{1}{\sqrt{512}} H_{512}\,(s \odot x)$$

- $s \in \{-1, +1\}^{512}$ is a deterministic Rademacher sign vector generated via `SplitMix64` from fixed seed RabitqSeed $=$ 0x9E3779B97F4A7C15, and $s \odot x$ denotes element-wise multiplication $s_i x_i$.
- H_{512} is the in-place Fast Walsh-Hadamard Transform (`FHT512`), computed in $512 \log_2(512) = 4\,608$ butterfly add/sub operations, scaled by $\frac{1}{\sqrt{512}} \approx 0.044194174$.
- Because $Q = \frac{1}{\sqrt{512}} H_{512}\,\mathrm{diag}(s)$ satisfies $Q^\top Q = I_{512}$, the mapping is an **exact Euclidean isometry** ($\|p\|_2 = \|x\|_2$, $\langle p, p' \rangle = \langle x, x' \rangle$) while spreading coordinate variance evenly across all `512` dimensions.

2.2 Coordinate-Mean Centering and 1-Bit Sign Encoding (`QuantizeRabitq`)

The rotated vector $p \in \mathbb{R}^{512}$ is centered by its coordinate arithmetic mean prior to 1-bit sign encoding:

$$\overline{p} = \frac{1}{512} \sum_{j=0}^{511} p_j, \qquad u_j = p_j - \overline{p}, \qquad c_j = \mathbf{1}_{u_j \geq 0} = \begin{cases} 1 \text{ if } p_j - \overline{p} \geq 0 \\ 0 \text{ if } p_j - \overline{p} < 0 \end{cases}$$

The `512` sign bits c_j are packed little-endian into `8 uint64` words via $\mathrm{Bits}\left[\left\lfloor \frac{j}{64} \right\rfloor\right] \leftarrow \mathrm{Bits}\left[\left\lfloor \frac{j}{64} \right\rfloor\right] \text{ OR } (1 \ll (j \bmod 64))$. When continuous Euclidean distance estimation is needed (`RabitqCode`), the centered norms $L_1 = \sum_{j=0}^{511} |u_j|$ and $L_2 = \|u\|_2$ link Hamming distance $d_H(q, c)$ to squared Euclidean distance via the angular cosine estimator (`AsymmetricDistanceCodes`):

$$\cos\hat{\theta}(q, c) = 1 - \frac{2\, d_H(q, c)}{512}, \qquad \hat{d}_2^2(q, c) = \|u_q\|_2^2 + \|u_c\|_2^2 - 2\,\|u_q\|_2\,\|u_c\|_2 \cos\hat{\theta}(q, c)$$

(An asymmetric 1-bit × 5-bit transposed bit-plane estimator without lookup tables is also provided in `BitProductScalar`: $\hat{d}^2_{\mathrm{asym}} = \|q\|_2^2 + \|u_c\|_2^2 - 2\|u_c\|_2^2 \langle s_c, q \rangle / \|u_c\|_1$, with $s_c \in \{-1, +1\}^{512}$.)

2.3 Unrolled Hamming Distance, `L1a`/`L1b` Arbitration, and Calibrated Radius R_{block}

At runtime (`SearchCentroid`), query bitcode q is compared against each stored threat centroid $c \in \mathcal{C}$ using 8 hardware `POPCNT` instructions with early word-level termination once `maxHamming` is exceeded (`codebookHammingBounded`):

$$d_H(q, c) = \sum_{w=0}^{7} \mathrm{popcount}(q_w \oplus c_w) \in \{0, \ldots, 512\}, \qquad d_H^*(q) = \min_{c \in \mathcal{C}} d_H(q, c)$$

Defensive monotonicity across `L1a` and `L1b` (`inference_cascade.go`) enforces three geometric rules:

1. **Subordination of `L1a` Conformal Probe**: `RabitqProbe` computes mean normalized Hamming distances D_1, D_2 to up to `32` benign/hostile prototypes and measures class dispersion Spread_k relative to a 64-bit word-level majority centroid (`classSpread`: word w is `^uint64(0)` iff $2\sum_{i=1}^{n} \mathrm{popcount}(\mathrm{proto}_i[w]) \geq 64n$, else `0`). Even if `L1a` returns benign `ProbeConfirm`, early benign release is **forbidden** whenever $d_H^*(q) \leq \max(R_{\mathrm{block}}, 24)$ outside ontologically certified contexts (`ontoVerdict != OntoVerdictAllow`).
2. **Direct `L1b` Threat Veto**: if $d_H^*(q) \leq R_{\mathrm{block}}$, the engine issues `VerdictBlock` ($Q_8 = 2(128 - d_H^*(q))$), except on registrable apex DNS names lacking subdomains (`isApexDomain`), where `PredictINT8` is lazily evaluated so a conformally certified benign domain overrides lexical collisions.
3. **Suspicious Gray-Band Quarantine**: if $R_{\mathrm{block}} < d_H^*(q) \leq 24$ (and neither apex-certified benign nor ontologically allowed), the event is escalated at minimum to `VerdictQuarantine`.

During compilation (`CalibrateBlockRadius`), R_{block} is calibrated exclusively on benign training set B_{train} at empirical false-block rate $\leq \alpha_{\max} = 1\%$ (`benignBlockRateMax = 0.01`, bounded by `FloppyDefaultBlockRadius = 12` and floored at `1`):

$$R_{\mathrm{block}} = \max\left(\{1\} \cup \left\{ r \in \{0, \ldots, 12\} \;\middle|\; \frac{1}{|B_{\mathrm{train}}|} \sum_{x \in B_{\mathrm{train}}} \mathbf{1}_{d_H^*(q(x)) \leq r} \leq 0.01 \right\} \right)$$

3. Empirical Class Centroids and Quantized `INT8` Linear Head

Events reaching `L1b` without triggering a threat centroid veto are scored by an 8-bit signed linear head (`DecisionClass`). In `c2forge` (`computeEmpiricalHead`), the class weights implement an **isotropic Fisher / Nearest-Centroid linear discriminant** over real centered training embeddings B (benign, class `0`) and H (hostile, class `1`):

$$\mu_B = \frac{1}{|B|} \sum_{x \in B} x \in \mathbb{R}^{512}, \qquad \mu_H = \frac{1}{|H|} \sum_{x \in H} x \in \mathbb{R}^{512}$$

The `INT8` weights $W_0, W_1 \in [-127, 127]^{512}$ (with $\mathrm{Bias}_0 = \mathrm{Bias}_1 = 0$) quantize the empirical centroid difference:

$$W_0[d] = \mathrm{clamp}_{[-127,127]}(\mathrm{trunc}(127 \cdot (\mu_B[d] - \mu_H[d]))), \qquad W_1[d] = \mathrm{clamp}_{[-127,127]}(\mathrm{trunc}(127 \cdot (\mu_H[d] - \mu_B[d])))$$

At runtime (`PredictINT8`), after rejecting any input containing IEEE-754 `NaN` or `±Inf` coordinates, each coordinate is quantized to $q_d = \text{clamp}_{[-127,127]}(\text{trunc}(127 \cdot x_d))$, and the integer dot product and normalized score are computed:

$$S_c(x) = \text{Bias}_c + \sum_{d=0}^{511} q_d \, W_c[d], \qquad \tilde{S}_c(x) = \frac{S_c(x)}{127 \times 512} = \frac{S_c(x)}{65\,024}, \qquad c^* = \arg\max_{c \in \{0,1\}} S_c(x)$$

Prediction c^* is certified **conformal** iff $\tilde{S}_{c^*}(x) \geq \hat{q}_{c^*}$ (`QHat`), where `computeEmpiricalHead` sets $\hat{q}_c = 0.4 \times \overline{\tilde{S}}_c$ if mean normalized training score $\overline{\tilde{S}}_c > 0.02$ (else 0.05). In `L1b`, $c^* = 1$ with conformity yields `VerdictBlock`; without conformity, `VerdictQuarantine`.

4. Offline Recertification and Anti-Poisoning `CUSUM` Drift Guard

In production, `.c2book` floppies are immutable `PROT_READ` mappings authenticated by `HMAC-SHA256` (header + body) and Castagnoli `CRC32C`, swapped atomically in $O(1)$ via `FloppySlot`. Live traffic **never mutates centroids in place**.

During periodic offline recertification producing candidate reference centroids $\mu_0, \mu_1, \ldots, \mu_t \in \mathbb{R}^{512}$, `CusumDriftGuard.CheckUpdate` (`drift_guard.go`) prevents gradual **boiling-frog poisoning** by enforcing four cumulative gates (`DefaultDriftGuardConfig`):

1. **IEEE-754 Finiteness Gate**: immediate rejection if $\exists d \in \{0, \ldots, 511\}, \mu_t[d] \in \{\text{NaN}, \pm\infty\}$, preventing unordered float comparisons from bypassing thresholds.
2. **Instantaneous Euclidean Step Bound** ($r_{\max} = 15.0$):

$$\Delta_t = \|\mu_t - \mu_{t-1}\|_2 = \sqrt{\sum_{d=0}^{511} (\mu_t[d] - \mu_{t-1}[d])^2} \leq r_{\max}$$

3. **Total Cumulative Drift Budget** ($B = 50.0$):

$$D_t = \sum_{\tau=1}^{t} \Delta_\tau \leq B$$

4. **Page's One-Sided Sequential `CUSUM` Detector** (slack $k = 0.1$, alarm threshold $h = 5.0$, $S_0^+ = 0$):

$$S_t^+ = \max(0, S_{t-1}^+ + (\Delta_t - k)) \leq h$$

If any condition fails, candidate μ_t is rejected, `alertCount` is incremented, and the active reference centroid stays pinned to μ_{t-1}.